Adapting Astronet vetting to TGLC — strategy memo

Astronet / TGLC adaptation · prepared for Pablo

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1. The question

We want the production Astronet **vetting** ensemble (`pablomer_final`, 10 models, trained Mar-2026 on **QLP** light curves) to work on **TGLC** (TESS–Gaia Light Curve) photometry. The near-term production plan is a **hybrid light curve**: keep the already-preprocessed **QLP for old sectors**, add **TGLC for new sectors**, stitch, then phase-fold.

Two options were put on the table:

1. **Mimic the new preprocessing in our training set**, then retrain.
2. **Take the trained model and fine-tune on a few recent TGLC examples — and that's it.**

Short answer: neither pure option is right, but they are closer than they look. The correct plan is **Option 1's goal (make the training distribution match what the model sees at inference) achieved through Option 2's mechanism (warm-start *continue-training* the existing ensemble)** — applied to a *representative* mixed-source training set rather than a handful of examples, and staged so we are not blocked waiting on QLP. The rest of this memo explains why.

2. The one rule that decides everything

This codebase has already paid for the central lesson, in production (`Astronet-Triage/DEPLOYMENT.md`, gotcha #1):

> **Train and inference preprocessing MUST match.**

When the *only* mismatch was regular- vs scatter-weighted **binning** (a far smaller change than swapping the entire photometry pipeline), **24–31 % of top-class predictions flipped** on sector 96, and the decision threshold had to move from 0.75 toward ~0.9 on sector 97. QLP→TGLC changes the *photometry itself* — noise floor, systematics, detrending residuals, and, most dangerously for vetting, **apparent transit depths and blended-EB signatures** (TGLC decontaminates blends using Gaia).

The direct consequence: **running the QLP-trained model on TGLC unchanged is not viable.** It will be wrong in ways that are silent and biased toward exactly the new objects we care about. So *both* options agree we must touch training. The real debate is only about **how** and **how much**.

3. The data reality (why we can't just "do Option 1 fully")

From the June-2026 status:

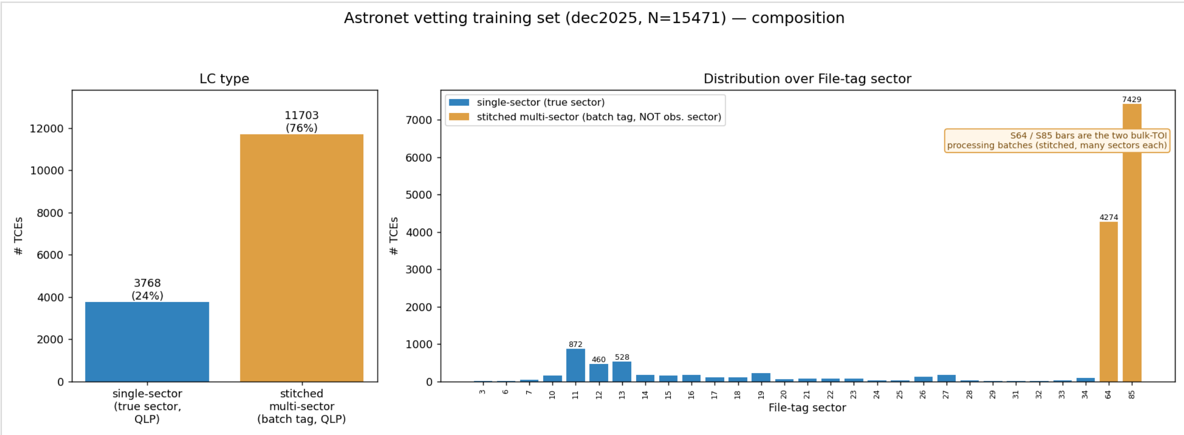
- Te Han has processed the **main mission + EM1 (≤ S56)** as TGLC — **but without QLP quality flags and without QLP detrending.** This is "raw" TGLC, *not* pipeline-consistent with how the model was trained.
- The **S56–S93 reprocessing through QLP** (TGLC fluxes run through QLP flags + detrending) is **underway but blocked**: cutouts are done; PSF and downstream steps are stuck on QLP being **overloaded on I/O**.
- Therefore a **uniformly pipeline-consistent TGLC training set across all sectors does not exist yet.** What we realistically have today is Te's raw ≤ S56.

So a faithful, from-scratch "all-TGLC" retrain (the maximalist reading of Option 1) is **not buildable right**

now. Any plan has to be robust to the QLP blockage.

4. What the current training set actually looks like (and a key surprise)

This shapes the decision more than anything. The labeled set behind the production model (dec2025) is **N = 15,471 TCEs — Planet 5,823 · EB 3,390 · Junk 6,258**. But the headline is *what kind of light curve each example is*:



Training-set composition: single-sector vs stitched, and File-tag sector

- **~76 % of the training set (11,703 TCEs) are MULTI-SECTOR *stitched* QLP light curves** — the two bulk-TOI batches. Their filenames say `s0064` / `s0085`, but **that is a processing/batch tag, not an observation sector**: these files have a median time span of **~760 days** (many sectors stitched together). Only the remaining **24 % (3,768, the `mk_*` files) are genuine single-sector** LCs (median span ~27 d) tagged with their *true* sector S3–S34.
- **This corrects a natural assumption:** although `read_and_process_light_curve()` opens *one* FITS per TCE, that one FITS is usually a stitched multi-sector curve. **So the model is already trained, predominantly, on stitched multi-sector QLP.** The production plan to "combine QLP-old + TGLC-new into one light curve, then fold" is therefore the **same paradigm we already use** — *not* a new capability. The adaptation is simply to **swap the per-sector photometry source inside the existing stitch.** That is very good news: less to build, and train==inference is easier to preserve.
- Camera/CCD is only well-defined for the single-sector subset (a stitched LC spans several cam/CCDs). On those 3,768 it resolves 100 % (camera 1>2>3>4; CCD 2 most populated) — see `figures/training_set_camccd_singlesector.png`.
- The batch labels are also class-segregated: the **s64 batch is ≈ all Planets** (4,257 P) and the **s85 batch is ≈ all Junk** (6,053 J) plus most recent Planets — see `figures/training_set_sector_per_label.png`.

Why this is the crux. Because every stitched example *already* mixes many sectors, "make the recent sectors TGLC" (constraint 3b) becomes a **per-segment source mix *within* each example** — each example becomes a QLP+TGLC blend. That **naturally defuses** the worst confound (a whole example being purely-one-source acting as a label proxy). What remains to watch is the *batch*-level segregation (s64≈Planet, s85≈Junk were processed separately) when choosing which segments get TGLC. Constraint 3a ("use the ≤ S56 TGLC") would replace the *old-sector portions* of these stitched curves — but that TGLC is "raw" (no QLP flags/detrend), so its usability is unproven. Neither is free — which is exactly why **Phase 0 (measure the gap) comes first.**

5. The two options, evaluated

Option 2 as literally stated — "fine-tune on a few recent TGLC examples, done"

Weakest option.

- A handful of examples cannot close a *whole-distribution* photometric shift; it only nudges the decision boundary near those points.
- High **catastrophic-forgetting** risk against 15 k QLP examples.
- The recent-sector TGLC examples are themselves **class- and sector-imbalanced** (mostly recent TOIs), reinforcing the "source = label" confound above.
- If production feeds a **hybrid** (QLP-old + TGLC-new) curve, a model tuned only on *pure* recent TGLC still doesn't match the input it will see.

But the mechanism it points at — warm-start fine-tuning rather than from-scratch — is correct. The vetting model already warm-starts from a frozen triage backbone, so continue-training is natural and cheap. The flaw is "a few examples," not "fine-tune."

Option 1 as literally stated — "mimic preprocessing, retrain"

Right principle, currently infeasible in full. It is the only option that respects the train==inference rule. But (§3) we can't assemble a pipeline-consistent all-TGLC set yet, and a full from-scratch retrain throws away the validated, hard-won ensemble behavior (and its threshold calibration) for no added benefit over continue-training.

6. Recommendation — Option 1's goal via Option 2's mechanism, in three phases

> **Build the smallest training set whose *per-example source mix matches what production will feed the model*, then warm-start continue-train the existing ensemble on it, and recalibrate the threshold.** Do it in phases so Phase 0 delivers value even while QLP is blocked.

Phase 0 — Characterize the gap before changing the model (do this now; cheap; unblocked).
This *is* the "Pablo + David: is the ≤ S56 TGLC usable?" task, made quantitative.

- Pick targets that have **both** QLP and TGLC in the **same** sector (Te's ≤ S56 TGLC vs our existing QLP for the same TICs). Run the **current** model on each and measure prediction drift: Δdisp_p, top-class flip rate, broken down by **class, magnitude, and crowding**.
- Produce **view-level diffs** (global/local/secondary overlays, QLP vs TGLC) to see *where* TGLC differs — depth? scatter? secondary? odd/even?
- Reuse existing machinery: the s96/s97 comparison harness (`/pdo/users/pablomer/vetting_comparison_s96cam1/` , `…_s97/`) and `Astronet-Triage/rank_view_differences.py` .
- **Deliverable / decision:** a number for "how far is TGLC from QLP, per class," and a yes/no on whether raw ≤ S56 TGLC is usable *as-is* or needs QLP-equivalent flagging/detrending first.

Phase 1 — Build a training set that matches production — *inside the existing stitch*.

- **Keep the stitched multi-sector paradigm.** Since ~76 % of training is already stitched multi-sector QLP (§4), and production will also be stitched, the move is **not** to go single-sector (that would *break* train==inference). Instead, modify the stitcher / h5→FITS converter so that **chosen (recent) sectors are pulled from TGLC instead of QLP** within the same stitched curve — then run it through the **same** `generate_input_records_3.py` (identical binning + scatter weighting). Each training example becomes a QLP+TGLC blend whose source mix mirrors what production will feed the model.
- Keep QLP for sectors where TGLC is unavailable; the result is a **mixed-source** set with the production source mix baked in per example.
- **Make the TGLC inputs pipeline-consistent**: if Phase 0 shows raw TGLC's missing QLP flags/detrending hurt, either wait for the QLP-pipeline TGLC for those sectors, or apply our own flag+detrend step so TGLC enters preprocessing the same way QLP does.
- **Mind the batch-level confound** (§4): the per-segment blend already defuses the worst whole-example "source = label" trap, but keep the s64≈Planet / s85≈Junk batch segregation in mind when choosing which segments get TGLC; balance sampling if needed.

Phase 2 — Warm-start continue-train + recalibrate.

- Initialize from `pablomer_final` weights and continue-train at a **low learning rate** on the mixed-source set, **keeping QLP examples in the mix** to prevent forgetting. (This is Option 2's mechanism, but on a representative set — not a handful.)
- **Recalibrate the decision threshold** on validation for the new source mix. Every preprocessing/source change has moved the operating point (s97 evidence); TGLC will move it again. Recalibrate from val data, **never** from a contentious live subset.
- Validate with the standard eval (PR curves, per-class recall) **and** a fresh QLP-vs-TGLC behavior comparison on a held-out sector.

Why this beats both pure options: it closes the actual distribution gap (unlike a few-example tune), it does not require an all-TGLC set that doesn't exist (unlike full Option 1), it preserves the validated ensemble and is far cheaper than from-scratch, and Phase 0 produces decision-useful results *even while QLP is blocked*.

7. Decision gates (what flips the plan)

Gate	Question	If "yes / large"	If "no / small"
G1	Is raw ≤ S56 TGLC usable without QLP flags/detrend? (Phase 0)	use it directly for Phase 1	add our own flag+detrend, or wait on QLP-pipeline TGLC
G2	Which sectors get TGLC inside the stitch? (production *and* training stay stitched — §4)	more TGLC segments → broader coverage, watch batch confound	recent-only TGLC → smaller, safer change
G3	Is the TGLC→QLP prediction shift large? (Phase 0)	broader TGLC coverage + per-class attention + threshold recal	a light warm-start fine-tune is enough
G4	QLP reprocessing ETA (S56–S93)?	gates pipeline-consistent new-sector TGLC	proceed on ≤ S56 now

8. Risks to watch

- **EB depths via decontamination.** TGLC changes apparent depths and can erase blended-EB signatures → the planet-vs-EB boundary moves. **Watch EB recall specifically.**
- **Missing quality flags in raw TGLC.** Unmasked momentum-dump / scattered-light cadences → spurious transit-like or secondary features → false positives. Must flag before preprocessing.
- **Batch-as-label shortcut** (§4) — the s64≈Planet / s85≈Junk *processing batches* are class-segregated; the per-segment QLP/TGLC blend helps, but don't let "which batch got TGLC" become a label proxy.
- **Threshold drift** — assume the operating point moves; budget a recalibration every time the source mix changes.

9. Immediate next actions

1. **Phase 0 now** (unblocked): assemble the QLP∩TGLC same-sector overlap set; run the current model on both; produce drift stats + view diffs. *(Pablo + David.)*
2. **Confirm with Willie/Glen** the QLP reprocessing status + ETA (gate G4).
3. **Decide which sectors get TGLC inside the stitch** (gate G2) — recommend recent-sectors-only first; keep the stitched multi-sector paradigm (it already matches training, §4).
4. Only then build the mixed-source training set (Phase 1) and warm-start fine-tune (Phase 2).

Companion material: running notes and data inventory in `TGLC_adaptation/NOTES.md`; training-set distribution figures in `TGLC_adaptation/figures/`.